

1(a)	units of F : kg ms^{-2}	C1
	units of ρ : kg m^{-3} and units of v : ms^{-1}	C1
	units of K : $\text{kg ms}^{-2} / [\text{kg m}^{-3} (\text{ms}^{-1})^2]$ $= \text{m}^2$	A1
1(b)(i)	$K\rho = 1.5 / 33^2$	C1
	$= 1.38 \times 10^{-3}$ $F_D = 1.38 \times 10^{-3} \times 25^2$ or $F_D / 1.5 = 25^2 / 33^2$ $F_D = 0.86 \text{ N}$	A1
1(b)(ii)	$a = (1.5 - 0.86) / (1.5 / 9.81)$ or $a = 9.81 - [0.86 / (1.5 / 9.81)]$	C1
	$a = 4.2 \text{ ms}^{-2}$	A1
1(c)	initial acceleration is $g/9.81 (\text{ms}^{-2})$ /acceleration of free fall	B1
	acceleration decreases	B1
	final acceleration is zero	B1